

are more likely to not complete the order and cancel shares. This results in the actual measured costs being lower than they would have been had it not been for the price momentum and market conditions. Unfortunately we do not have a full audit trail in this case to be able to incorporate opportunistic trading and opportunity cost into our error analysis. But investors with a full audit trail will be equipped to properly incorporate opportunity cost as well as survivorship bias and opportunistic trading.

The next question is how far off the estimated cost is from the actual cost for the larger orders. Especially considering some orders are exposed to much greater market risk than others. To address this question we computed the z-score for each of the orders. That is,

$$Z = \frac{\text{Actual Cost} - \text{Estimated MI}}{\text{Timing Risk}} \quad (5.18)$$

The estimated timing risk is computed as:

$$TR = \sigma \cdot \sqrt{\frac{1}{3} \cdot \frac{1}{250} \cdot \text{Size} \cdot \frac{1 - POV}{POV}} \cdot 10^4 bp \quad (5.19)$$

Regardless of the distribution of the error, if the model is accurate the mean z-score will be zero and the variance will be one. That is,

$$Z \sim (0, 1)$$

We computed the average z-score in each size bucket. This is shown in [Figure 5.8d](#). This analysis shows that the risk adjusted error is not as inaccurate as it first appears. The average z-score for all order sizes, while significantly different from zero, is still within  $\pm 1$  standard deviation. To be more exact, the z-score is  $\pm 0.25$  standard units for sizes up to 30% ADV. Thus, while the model is overestimating actual trading costs (likely due to opportunistic trading and survivorship bias), the risk adjusted error term is not considered grossly erroneous. The error is quite reasonable and thus not a large concern.

### Stock Specific Error Analysis

The next step in our error analysis is to determine if there is anything specific to the stock or particular company that would help improve the accuracy of the model and reduce estimation error. For this step we follow the techniques presented by Breen, Hodrick and Korajczyk (2002).

Our error analysis was carried out by estimating the expected market impact cost using the parameters determined above and comparing these estimates to the actual order costs. The average error measured as the estimated cost